

Due Date: Wednesday, October 7 at 11:59 PM

Overview: In this assignment, you'll practice working with a private API by connecting an application to the Spotify Web API. The goal of this assignment is to help you understand how APIs work, how to make requests to an API, how to work with JSON data, and how to use JS to dynamically update a webpage.

Your task: You'll retrieve information about an artist, their albums, and their tracks, then use JS to display that information on a provided project.

Requirements:

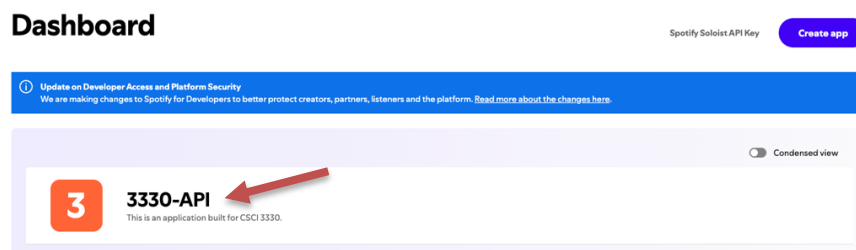
1. Create a Spotify Developer App. Before you can make requests to the Spotify API, you'll need to create a Spotify Developer application and obtain an access token. Follow the Spotify's instructions in the link below to create a Spotify Developer app.

<https://developer.spotify.com/documentation/web-api>

NOTE: When you create the Spotify app, you'll be required to enter information about the app. You may use:

- App Name: **My App**
- App Description: **This is my first Spotify app**
- Redirect URI: **http://127.0.0.1:3000**
- Which API/SDKs are you planning to use? **Web API**

After your app is created, you can visit your dashboard to obtain your client ID and Client Secret. Click in the app to see more details (Note that my app was named 3330-API).



You should see a similar screen as the one in the screenshot below. Take a note of the Client ID and click on the link “view client secret” to access this string.

Dashboard > 3330-API Metrics

3 Basic Information

Basic Information User Management

Client ID
b425621ec3df49d7a57efc6c58d510

App Status
Development mode

Refresh Token Lifetime 180 days
[View client secret](#)

App name
3330-API

App description
This is an application built for CSCI 3330.

2. Create your Spotify's access token. There are many ways to generate an access token. However, I recommend you to use Postman.

- a. Make a POST request to: <https://accounts.spotify.com/api/token>

POST ▼ <https://accounts.spotify.com/api/token> Send ▼

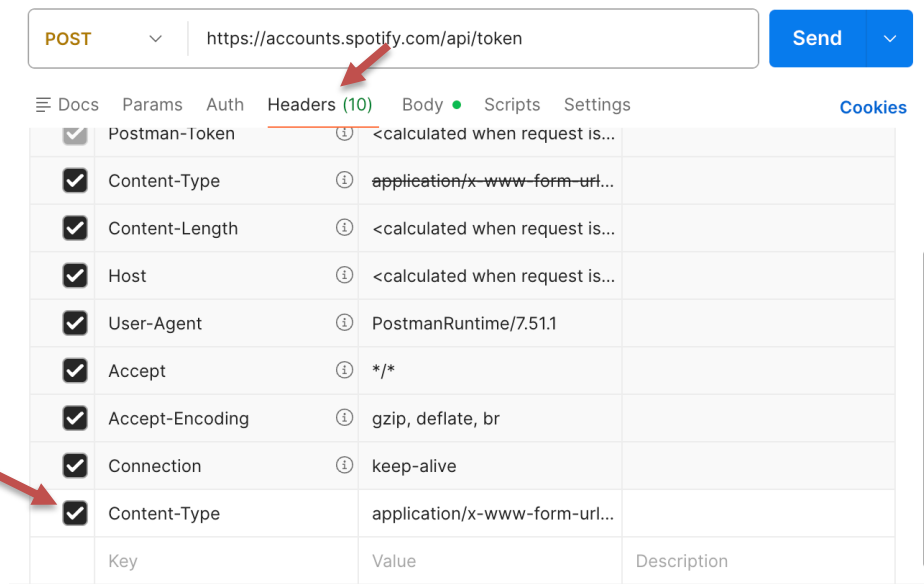
Docs Params Auth Headers (10) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

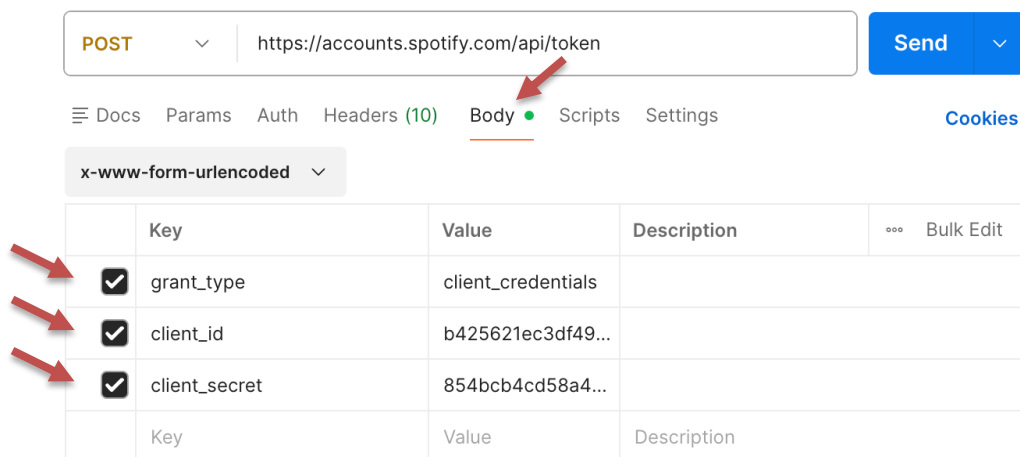
- b. Before you click Send, switch to the Headers tab and include:

Key	Value
Content-Type	application/x-www-form-urlencoded



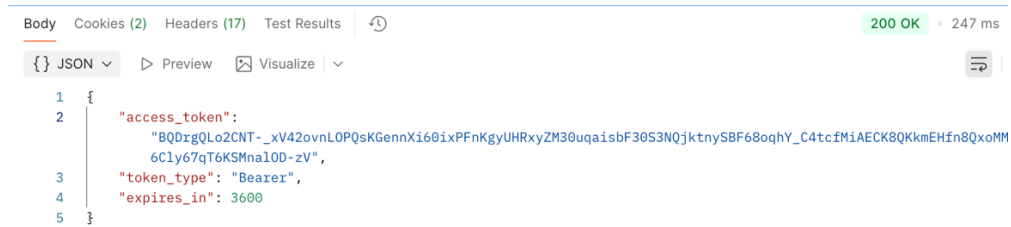
c. Also, switch to the Body tab and include three more parameters:

Key	Value
grant_type	client_credentials
client_id	<your-client-id >
client_secret	<your-client-secret>



d. Now you can click Send

e. Take a note of the access token. Postman will show your access token as a JSON object in the response.

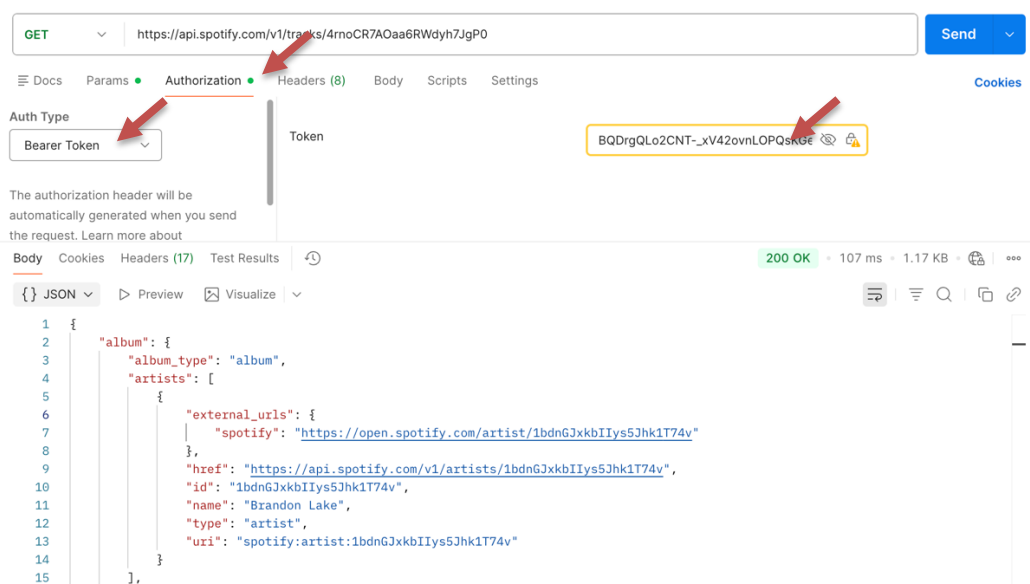


```
1 {
2   "access_token":
3     "BQDrgQLo2CNT-_xV42ovnLOPQsKGennXi60ixPFnKgyUHRxyZM30uqaisbF30S3N0jktnySBF68oqhY_C4tcfMiAECK8QKkmEHfn8QxoMM
4     6Cly67qT6KSMna10D-zV",
5   "token_type": "Bearer",
6   "expires_in": 3600
7 }
```

Important: Your Spotify access token is only valid for 1 hour (3600 seconds). When your token expires, you'll receive an authorization error, but don't worry, you can repeat these steps as many times are needed, simply send another POST request with your credentials.

Now, we can use the Spotify API.

3. Explore the Spotify API before working on your website by using Postman. Use the appropriate API endpoint for each GET request. Since this is a private API, you'll need to include your Spotify's access token in each request. To do so, switch to the Authorization tab in Postman, and select Bearer Token as the Authentication Type. This will allow you to enter your token as shown in the screenshot below. Lastly, click Send.



```
1 {
2   "album": {
3     "album_type": "album",
4     "artists": [
5       {
6         "external_urls": {
7           "spotify": "https://open.spotify.com/artist/1bdnGJxkbIIys5Jhk1T74v"
8         },
9         "href": "https://api.spotify.com/v1/artists/1bdnGJxkbIIys5Jhk1T74v",
10        "id": "1bdnGJxkbIIys5Jhk1T74v",
11        "name": "Brandon Lake",
12        "type": "artist",
13        "uri": "spotify:artist:1bdnGJxkbIIys5Jhk1T74v"
14      }
15    ]
16  }
```

If all is good, the response will return JSON data. Pay attention to how the JSON object is structured.

You will likely use the following endpoints:

- a. Get Artist. Use this endpoint to retrieve information about an artist:

<https://api.spotify.com/v1/artists/{id}>

Where to find the artist's id? Go to the Spotify website and search for the artist. On the artist's homepage, you can find their unique ID in the URL. For example, when searching for Brandon Lake, the artist's homepage displays their ID (highlighted in red) in the URL as follows:

<https://open.spotify.com/artist/1bdnGJxkbllys5Jhk1T74v>

more details: <https://developer.spotify.com/documentation/web-api/reference/get-an-artist>

- b. Get Album. Use this endpoint to retrieve information about an album. You'll need to go to the Spotify website and search for the album you're looking for. The URL will show you the album id.

<https://api.spotify.com/v1/albums/{id}>

more details: <https://developer.spotify.com/documentation/web-api/reference/get-an-album>

- c. Get Track. Use this endpoint to retrieve information about an individual track:

<https://api.spotify.com/v1/tracks/{id}>

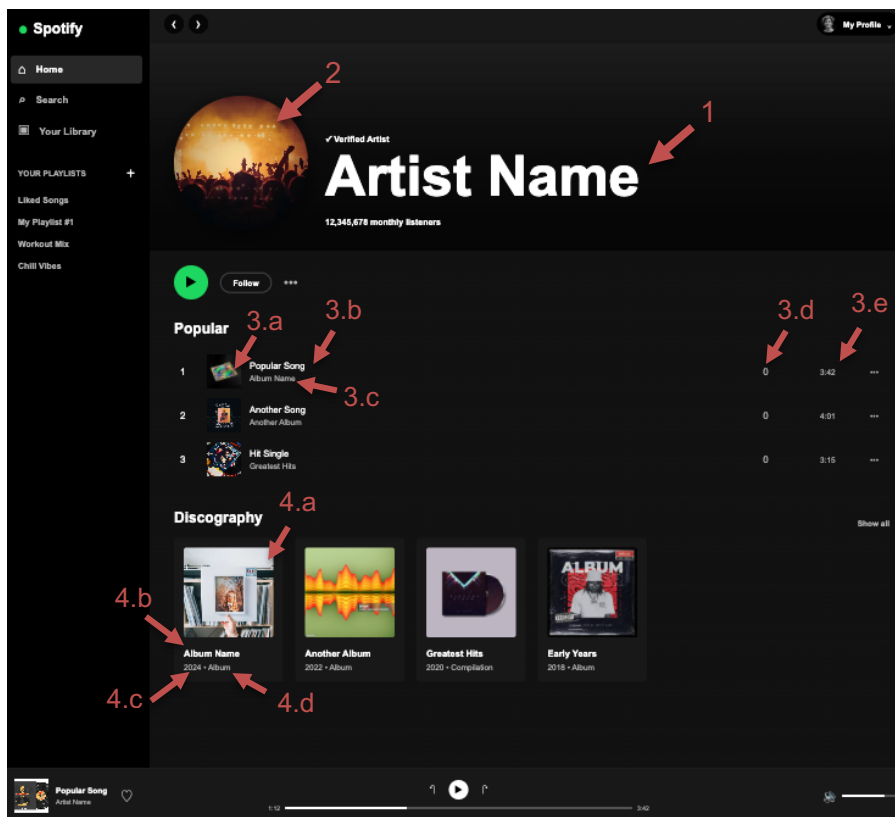
more details:

<https://developer.spotify.com/documentation/web-api/reference/get-track>

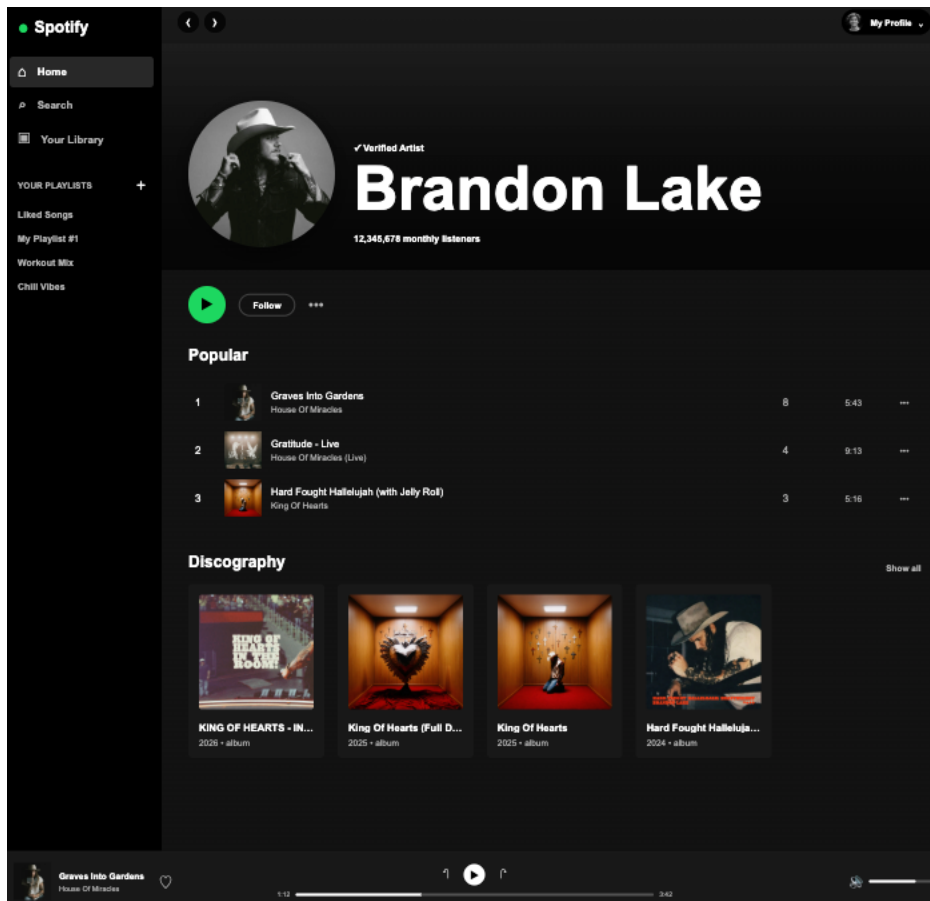
4. Use different API endpoint to retrieve data and update a webpage dynamically. A ZIP folder with a starter project is provided to you on Brightspace. Focus development on the JS file. You should **not change, add, or remove HTML elements** in the starter project. Instead, use JS to find the existing HTML elements and update their contents.

For this part of the assignment, you can select your favorite artist, songs and albums. Here's a list of the elements that should be dynamically populated using data from the various Spotify API endpoints:

1. Change the **artist's** name.
2. Change the **artist's** image.
3. In the Popular section, update each of the 3 **tracks**:
 - a. track album artwork
 - b. track title
 - c. track album name
 - d. track number and
 - e. track duration
4. In the Discography section, change each of the 4 **albums**:
 - a. album artwork
 - b. album name
 - c. album release date (must be formatted by using the `convertMsToMinSec()` function)
 - d. album type



UI BEFORE the implementation of API calls.



UI **AFTER** the implementation of API calls.

Hint: Use methods such as:

- `document.querySelector()`
- `document.querySelectorAll()`
- `element.textContent`
- `element.src`
- the `fetchAPI`:

```
const response = await fetch(
  `https://api.spotify.com/v1/artists/${artistId}`,
  {
    headers: {
      Authorization: `Bearer ${accessToken}`
    }
  }
);

const artist = await response.json();
```

Submission: Your solution must be hosted in GitHub, and **your repository should contain multiple commits** demonstrating your progress throughout the assignment. Use the Brightspace submission tools to submit a single text file containing the link to your GitHub repository.